Import The Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Read The Dataset

In [2]:
```
df=pd.read_csv(r"\User\Downloads\Pakistan Largest Ecommerce Dataset.csv",low_mem
```

In [3]:
```
df
```

Out[3]:

	item_id	status	created_at	sku	price	qty_ordered	gran
0	211131.0	complete	7/1/2016	kreations_YI 06-L	1950.0	1.0	
1	211133.0	canceled	7/1/2016	kcc_Buy 2 Frey Air Freshener & Get 1 Kasual Bo...	240.0	1.0	
2	211134.0	canceled	7/1/2016	Ego_UP0017-999-MR0	2450.0	1.0	
3	211135.0	complete	7/1/2016	kcc_krone deal	360.0	1.0	
4	211136.0	order_refunded	7/1/2016	BK7010400AG	555.0	2.0	
...	...	...	...	...	...	...	
1048570	NaN	NaN	NaN	NaN	NaN	NaN	
1048571	NaN	NaN	NaN	NaN	NaN	NaN	
1048572	NaN	NaN	NaN	NaN	NaN	NaN	
1048573	NaN	NaN	NaN	NaN	NaN	NaN	
1048574	NaN	NaN	NaN	NaN	NaN	NaN	

1048575 rows × 26 columns

In [4]:
```
df = df.dropna(how='all')
```

In [5]:
```
df = df.loc[:, ~df.columns.str.contains('^Unnamed')]
```

In [6]:
```
df.shape
```

Out[6]: (584524, 21)

Finding The Null Values

In [7]:
```
df.isnull().sum()
```

Out[7]:
```
item_id                   0
status                   15
created_at                0
sku                      20
price                     0
qty_ordered               0
grand_total               0
increment_id              0
category_name_1         164
sales_commission_code    137178
discount_amount           0
payment_method            0
Working Date              0
BI Status                 0
 MV                       0
Year                      0
Month                     0
Customer Since           11
M-Y                       0
FY                        0
Customer ID              11
dtype: int64
```

Drop And Fill The Null Values

In [8]:
```
df=df.dropna(subset=['status','sku','category_name_1','Customer Since','Customer
```

In [9]:
```
df['sales_commission_code']=df['sales_commission_code'].fillna('No code')
```

In [10]:
```
df.duplicated().sum()
```

Out[10]: np.int64(0)

Checking The Datatypes

In [11]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 584314 entries, 0 to 584523
Data columns (total 21 columns):
 #   Column                 Non-Null Count   Dtype  
---  ------                 --------------   -----  
 0   item_id                584314 non-null  float64
 1   status                 584314 non-null  str    
 2   created_at             584314 non-null  str    
 3   sku                    584314 non-null  str    
 4   price                  584314 non-null  float64
 5   qty_ordered            584314 non-null  float64
 6   grand_total            584314 non-null  float64
 7   increment_id           584314 non-null  str    
 8   category_name_1        584314 non-null  str    
 9   sales_commission_code  584314 non-null  str    
 10  discount_amount        584314 non-null  float64
 11  payment_method         584314 non-null  str    
 12  Working Date           584314 non-null  str    
 13  BI Status              584314 non-null  str    
 14   MV                    584314 non-null  str    
 15  Year                   584314 non-null  float64
 16  Month                  584314 non-null  float64
 17  Customer Since         584314 non-null  str    
 18  M-Y                    584314 non-null  str    
 19  FY                     584314 non-null  str    
 20  Customer ID            584314 non-null  float64
dtypes: float64(8), str(13)
memory usage: 98.1 MB
```

Fixing The Datatypes

In [12]:
```
df['created_at']=pd.to_datetime(df['created_at'])
df['item_id']=df['item_id'].astype('int')
df['qty_ordered']=df['qty_ordered'].astype('int')
df['Customer ID']=df['Customer ID'].astype('int')
df['Working Date']=pd.to_datetime(df['Working Date'])
df['Year']=df['Year'].astype('int')
df['Month']=df['Month'].astype(int)
```

In [13]:
```
df.columns = df.columns.str.strip()
```

In [14]:
```
df.rename(columns={
    'Customer ID': 'customer_id',
    'Customer Since': 'customer_since',
    'BI Status': 'bi_status',
    'Working Date': 'working_date'
}, inplace=True)
```

In [15]:
```
df.columns
```

Out[15]:
```
Index(['item_id', 'status', 'created_at', 'sku', 'price', 'qty_ordered',
       'grand_total', 'increment_id', 'category_name_1',
       'sales_commission_code', 'discount_amount', 'payment_method',
       'working_date', 'bi_status', 'MV', 'Year', 'Month', 'customer_since',
       'M-Y', 'FY', 'customer_id'],
      dtype='str')
```

In [16]:
```
df['status'] = df['status'].str.strip()
df['sku'] = df['sku'].str.strip()
df['payment_method'] = df['payment_method'].str.strip()
df['sales_commission_code'] = df['sales_commission_code'].str.strip()
df['bi_status'] = df['bi_status'].str.strip()
```

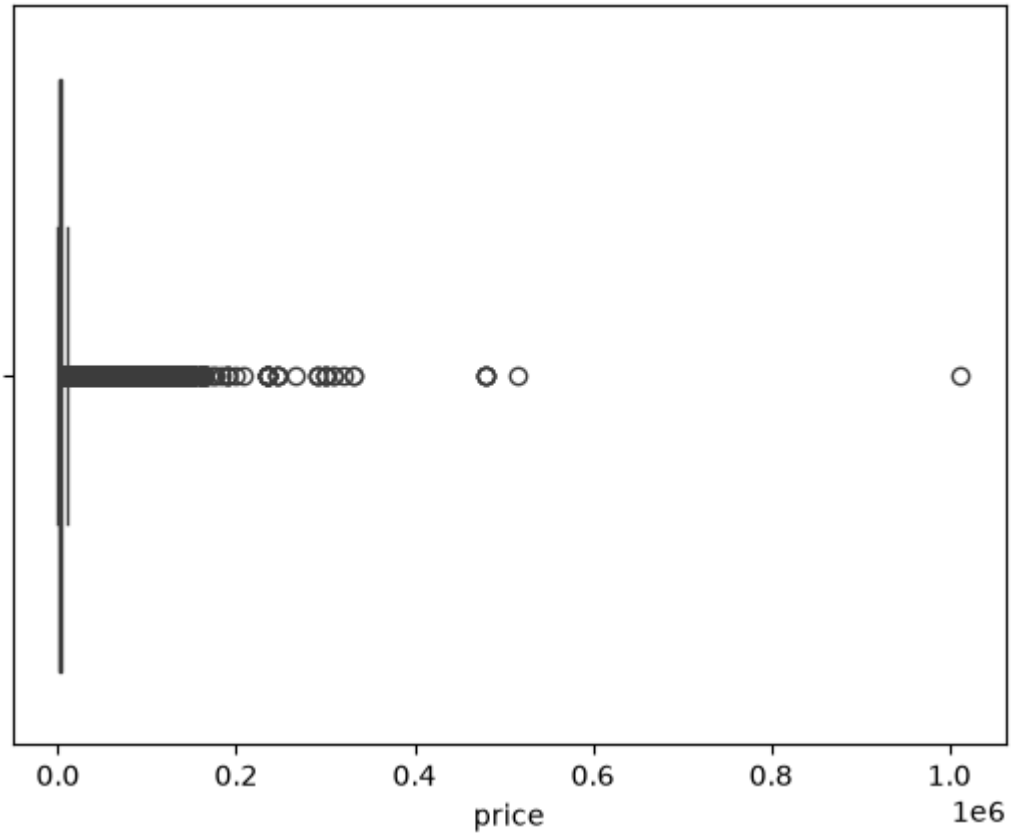
In [17]:
```
df = df[df['price'] > 0]
df = df[df['qty_ordered'] > 0]
```

In [18]:
```
df.shape
```

Out[18]: (582100, 21)

Checking The Outliers

In [19]:
```
sns.boxplot(x=df['price'])
plt.show()
```



In [20]:
```
print(df.nlargest(5, 'price')[['item_id', 'sku', 'price', 'category_name_1']])
```

```
        item_id                 sku      price    category_name_1
509081   821284  BAGGEM5A7038AC06C9A  1012625.9  Beauty & Grooming
309775   588634  ENTSAM59B7C1FA12478   515975.0      Entertainment
207276   470222      yamaha_EF12000E   479000.0         Appliances
207347   470297      yamaha_EF12000E   479000.0         Appliances
251158   521539      yamaha_EF12000E   479000.0         Appliances
```

Fix The Outliers

In [21]:
```
df = df[~((df['price'] > 1000000) & (df['category_name_1'] == 'Beauty & Grooming
```

In [22]:
```
df.shape
```

Out[22]: (582099, 21)

now load the data into pgadmin4

In [23]:
```
import pandas as pd
from sqlalchemy import create_engine
DB_USER = "postgres"
DB_PASS = "1242"
DB_HOST = "localhost"
DB_PORT = "5432"
DB_NAME = "Ecommerce"
engine = create_engine(f'postgresql://{DB_USER}:{DB_PASS}@{DB_HOST}:{DB_PORT}/{D
table_name = "data"
df.to_sql(
    name=table_name,
    con=engine,
    if_exists='replace',
    index=False
)
print("Data successfully transferred to PostgreSQL!")
```

```
Data successfully transferred to PostgreSQL!
```

In [ ]:
```
```